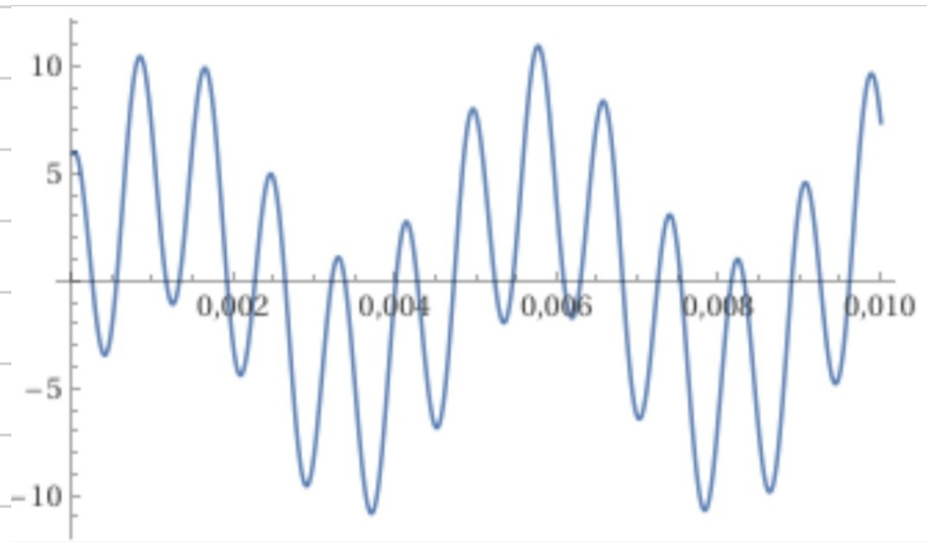


Ejercicio de Pulse Code Modulation (PCM)

$$v(t) = 6 \cdot \cos(2 \cdot \pi \cdot 1216 \cdot t) + 5 \cdot \sin(2 \pi \cdot 216 \cdot t)$$



1-

$$f_s = 1216 \text{ Hz}$$

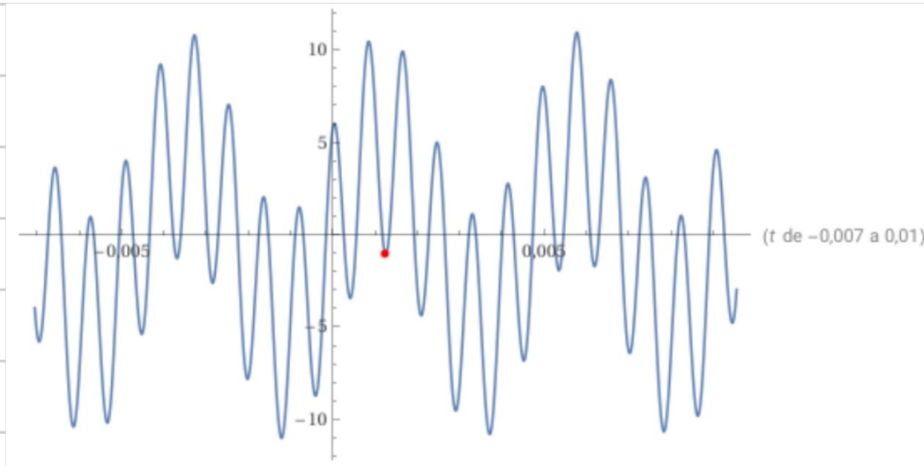
$$F_s = 2 \times f_{\text{max}} = 2432 \text{ Hz}$$

2- $V_p = \pm 11 \text{ V}$ $V_{pp} = 22 \text{ V}$

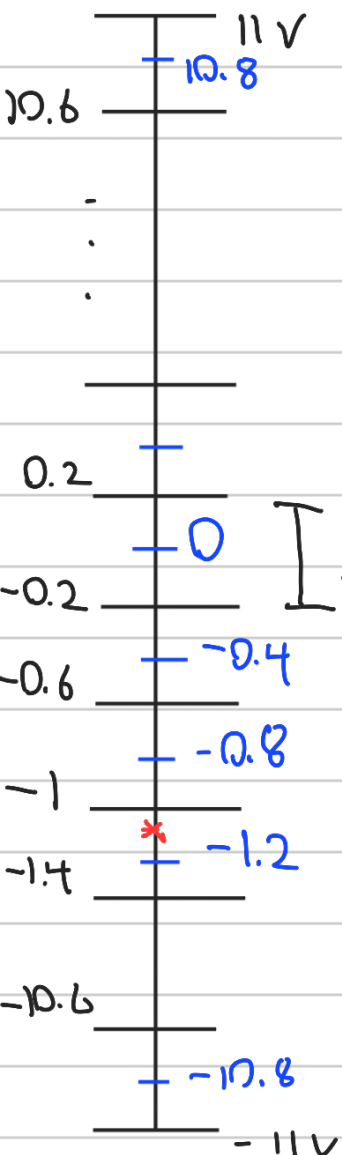
$$\Delta = 0.4 \quad \# \text{ niveles} = \frac{22 \text{ V}}{0.4} = 55$$

$$3- T_g = \frac{1}{F_g} = \frac{1}{2432 \text{ Hz}} = 411 \text{ ns}$$

$$k=3 \quad t = 3 \times T_g = 1233.55 \text{ ns}$$



$$V(1233.55 \text{ ns}) = -1.02667 \text{ V}$$



$\Delta = 0.4$ Rango # 28

Rango # 25

$$\text{Error de cuantización} = |\text{Real} - \text{Cuantizado}|$$

$$\text{Error} = |-1.02667 - (-1.2)| = 0.1733$$

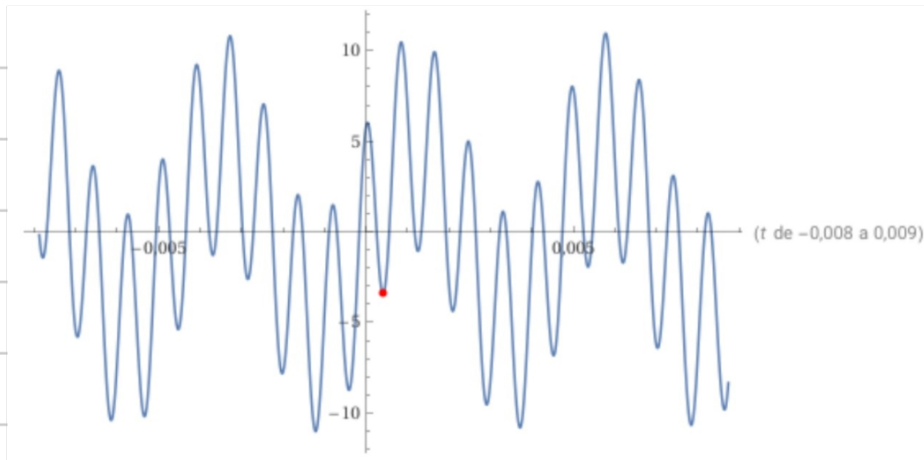
4-

$$b = \log_2 M = \log_2 55 = 5.78 = 6 \text{ bits}$$

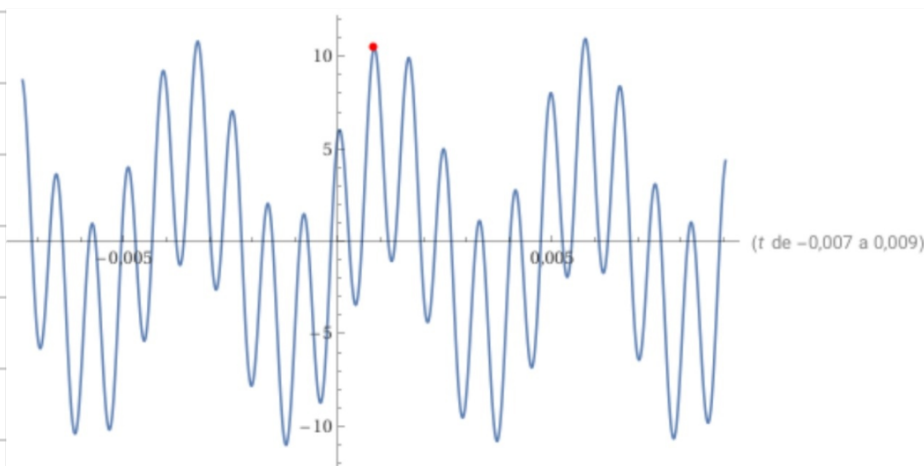
Desde 000000 hasta 110110

Rango #28 $\rightarrow$ 011100

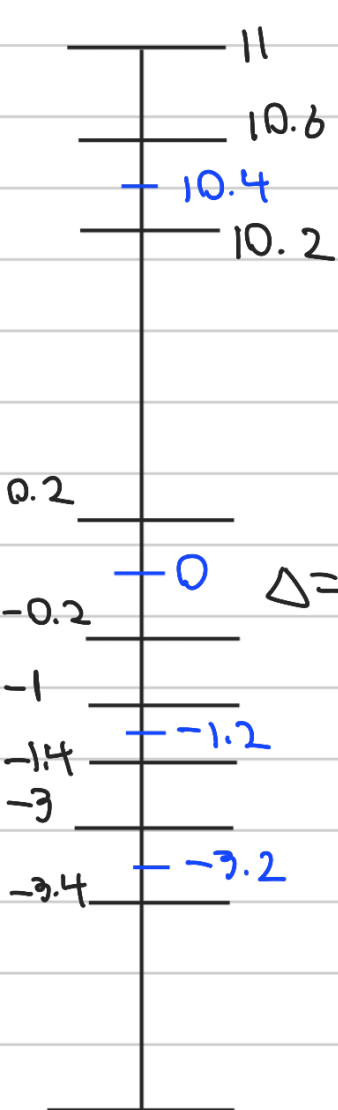
5-



$$V(411,18 \text{ ms}) = -3.35238 \text{ V}$$



$$V(822,36 \text{ ms}) = 10.4919 \text{ V}$$



Rango #55 $\rightarrow$ 110110

Rango #54 $\rightarrow$ 110101

0.2
-0.2 $\Delta = 0.4$ Rango #28

-1
-1.4 Rango #25 $\rightarrow$ 011000

-3
-3.4 Rango #20 $\rightarrow$ 010011

Salida del PCM

010011 110101 011000
 $\underbrace{\hspace{1.5cm}}_{k=1}$ $\underbrace{\hspace{1.5cm}}_{k=2}$ $\underbrace{\hspace{1.5cm}}_{k=3}$